Compression Gauge



Category: **Automotive** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Compressor gauge

What it is used for

To measure engine cylinder compression using a compatible compression gauge during automotive diagnosis. To compare readings with the vehicle manufacturer's specifications and identify possible mechanical problems.

Main parts

- Pressure gauge or digital display
- Pressure hose
- Spark-plug adapter or specified connector
- Release valve
- Fittings and seals

Before you start

1. Confirm the engine type and the correct gauge and adapter.
2. Review the vehicle service manual for the specified test procedure.
3. Park securely, apply the parking brake, and disable ignition and fuel delivery as directed.
4. Allow the engine to reach the specified test temperature if required.
5. Inspect the gauge, hose, fittings, and seals for damage.

How to use

1. Remove the specified spark plugs or access the designated test port according to the service manual.
2. Install the correct adapter securely without cross-threading.
3. Set the throttle and other conditions as specified by the manufacturer.
4. Crank the engine using the approved procedure and observe the gauge reading.
5. Repeat the measurement consistently for each cylinder.
6. Record the readings and compare them with specifications and cylinder-to-cylinder variation limits.
7. Release pressure, remove the gauge, and reinstall components to the specified torque.

⚠ Safety

- Keep hands, clothing, and tools away from moving engine parts.
- Follow the required ignition and fuel-system isolation procedure.
- Do not exceed the gauge or adapter pressure rating.
- Allow hot components to cool where required and wear eye protection.

Care & storage

- Release pressure before disconnecting the hose.
- Clean adapters and fittings after use.
- Inspect the hose, seals, and gauge for leaks or wear.
- Store the gauge in a protective case.

Common mistakes to avoid

- Using the wrong adapter or thread type.
- Failing to disable ignition or fuel delivery as specified.
- Comparing readings taken under different test conditions.
- Ignoring manufacturer limits and cylinder-to-cylinder differences.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar